

E-Bikes

Gao

2024-04-10

```
library(tidymodels)
```

```
## -- Attaching packages ----- tidymodels 1.1.1 --
```

```
## v broom      1.0.5      v recipes      1.0.10
## v dials      1.2.1      v rsample      1.2.0
## v dplyr      1.1.4      v tibble      3.2.1
## v ggplot2    3.5.0      v tidyr       1.3.1
## v infer      1.0.6      v tune        1.1.2
## v modeldata  1.3.0      v workflows   1.1.4
## v parsnip    1.2.0      v workflowsets 1.0.1
## v purrr      1.0.2      v yardstick   1.3.0
```

```
## -- Conflicts ----- tidymodels_conflicts() --
```

```
## x purrr::discard() masks scales::discard()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x recipes::step() masks stats::step()
## * Dig deeper into tidy modeling with R at https://www.tmw.org
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v forcats  1.0.0      v readr      2.1.5
## v lubridate 1.9.3      v stringr    1.5.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x readr::col_factor() masks scales::col_factor()
## x purrr::discard()    masks scales::discard()
## x dplyr::filter()     masks stats::filter()
## x stringr::fixed()    masks recipes::fixed()
## x dplyr::lag()        masks stats::lag()
## x readr::spec()       masks yardstick::spec()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(rpart.plot)
```

```
## Loading required package: rpart
##
```

```
## Attaching package: 'rpart'
##
## The following object is masked from 'package:dials':
##
##   prune
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following objects are masked from 'package:yardstick':
##
##   precision, recall, sensitivity, specificity
##
## The following object is masked from 'package:purrr':
##
##   lift
```

```
library(dplyr)
library(ggforce)
library(mctest)
library(olsrr)
```

```
##
## Attaching package: 'olsrr'
##
## The following object is masked from 'package:datasets':
##
##   rivers
```

```
library(jtools)
```

```
##
## Attaching package: 'jtools'
##
## The following object is masked from 'package:yardstick':
##
##   get_weights
```

```
library(ggcorrplot)
library(yardstick)
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:purrr':
##
```

```
##      some
##
## The following object is masked from 'package:dplyr':
##
##      recode
```

```
library(moments)
library(GGally)
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg      ggplot2
```

```
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following object is masked from 'package:car':
##
##      logit
##
## The following objects are masked from 'package:ggplot2':
##
##      %+%, alpha
##
## The following objects are masked from 'package:scales':
##
##      alpha, rescale
```

```
library(fastDummies)
```

```
## Thank you for using fastDummies!
## To acknowledge our work, please cite the package:
## Kaplan, J. & Schlegel, B. (2023). fastDummies: Fast Creation of Dummy (Binary) Columns and Rows from
```

```
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
##
## Attaching package: 'forecast'
##
## The following object is masked from 'package:yardstick':
##
##      accuracy
```

```
ebikes <- read.csv("b.csv")
ebikes
```

##	Year	US	USLin	USAge	USCar	USGas	PrivateMiles	PublicMiles
## 1	2012	70	3.486083	37.4	21819	3.62	5046	55.17
## 2	2013	159	5.253967	37.6	21751	3.51	5083	56.47
## 3	2014	193	5.788518	37.7	21640	3.36	5158	57.01
## 4	2015	130	4.750731	37.8	21365	2.43	5283	55.70
## 5	2016	152	5.137012	37.9	20815	2.14	5412	56.32
## 6	2017	263	6.757198	38.0	20004	2.42	5482	54.83
## 7	2018	369	8.003905	38.2	20077	2.72	5546	53.83
## 8	2019	423	8.569568	38.4	20278	2.60	5579	54.10
## 9	2020	416	8.498366	38.6	20935	2.17	4935	31.55
## 10	2021	750	11.410887	38.8	26500	3.01	5336	22.37
## 11	NA	NA	NA	NA	NA	NA	NA	NA
## 12	NA	NA	NA	NA	NA	NA	NA	NA
## 13	NA	NA	NA	NA	NA	NA	NA	NA
## 14	NA	NA	NA	NA	NA	NA	NA	NA
## 15	NA	NA	NA	NA	NA	NA	NA	NA
## 16	NA	NA	NA	NA	NA	NA	NA	NA

```
modellLin <- lm(USLin ~ USAge + USCar + USGas + PrivateMiles + PublicMiles, data = ebikes)
modellLin
```

```
##
## Call:
## lm(formula = USLin ~ USAge + USCar + USGas + PrivateMiles + PublicMiles,
##     data = ebikes)
##
## Coefficients:
## (Intercept)          USAge          USCar          USGas  PrivateMiles
## -2.183e+02    5.612e+00   -1.661e-05    1.435e+00    1.521e-03
## PublicMiles
## -1.565e-03
```

```
summary(modellLin)
```

```
##
## Call:
## lm(formula = USLin ~ USAge + USCar + USGas + PrivateMiles + PublicMiles,
##     data = ebikes)
##
## Residuals:
##      1      2      3      4      5      6      7      8
## -0.48906  0.25889  0.33246 -0.12838 -0.09143  0.44334  0.03928 -0.39169
##      9     10
## -0.01296  0.03954
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.183e+02  5.604e+01  -3.896   0.0176 *
## USAge        5.612e+00  1.494e+00   3.758   0.0198 *
## USCar       -1.661e-05  2.194e-04  -0.076   0.9433
## USGas        1.435e+00  4.353e-01   3.298   0.0300 *
## PrivateMiles 1.521e-03  1.624e-03   0.937   0.4019
## PublicMiles -1.565e-03  7.110e-02  -0.022   0.9835
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4458 on 4 degrees of freedom
## (6 observations deleted due to missingness)
## Multiple R-squared:  0.9841, Adjusted R-squared:  0.9643
## F-statistic: 49.6 on 5 and 4 DF, p-value: 0.001085

modell1 <- lm(US ~ USAge + USCar + USGas + PrivateMiles + PublicMiles, data = ebikes)
modell1
```

```
##
## Call:
## lm(formula = US ~ USAge + USCar + USGas + PrivateMiles + PublicMiles,
##     data = ebikes)
##
## Coefficients:
## (Intercept)      USAge      USCar      USGas PrivateMiles
## -1.491e+04    3.631e+02    6.071e-03    1.214e+02    2.131e-01
## PublicMiles
## -4.142e+00
```

```
summary(modell1)
```

```
##
## Call:
## lm(formula = US ~ USAge + USCar + USGas + PrivateMiles + PublicMiles,
##     data = ebikes)
##
## Residuals:
##      1      2      3      4      5      6      7      8
## -18.0671  9.5792 12.4074 -4.4096 -5.0970 19.4444 -1.8103 -12.9975
##      9     10
##  -0.6132  1.5637
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.491e+04  2.151e+03  -6.933  0.00227 **
## USAge        3.631e+02  5.732e+01   6.335  0.00318 **
## USCar        6.071e-03  8.419e-03   0.721  0.51078
## USGas        1.214e+02  1.670e+01   7.266  0.00191 **
## PrivateMiles 2.131e-01  6.232e-02   3.419  0.02681 *
## PublicMiles -4.142e+00  2.729e+00  -1.518  0.20362
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.11 on 4 degrees of freedom
## (6 observations deleted due to missingness)
## Multiple R-squared:  0.9969, Adjusted R-squared:  0.9929
## F-statistic: 253.2 on 5 and 4 DF, p-value: 4.326e-05
```

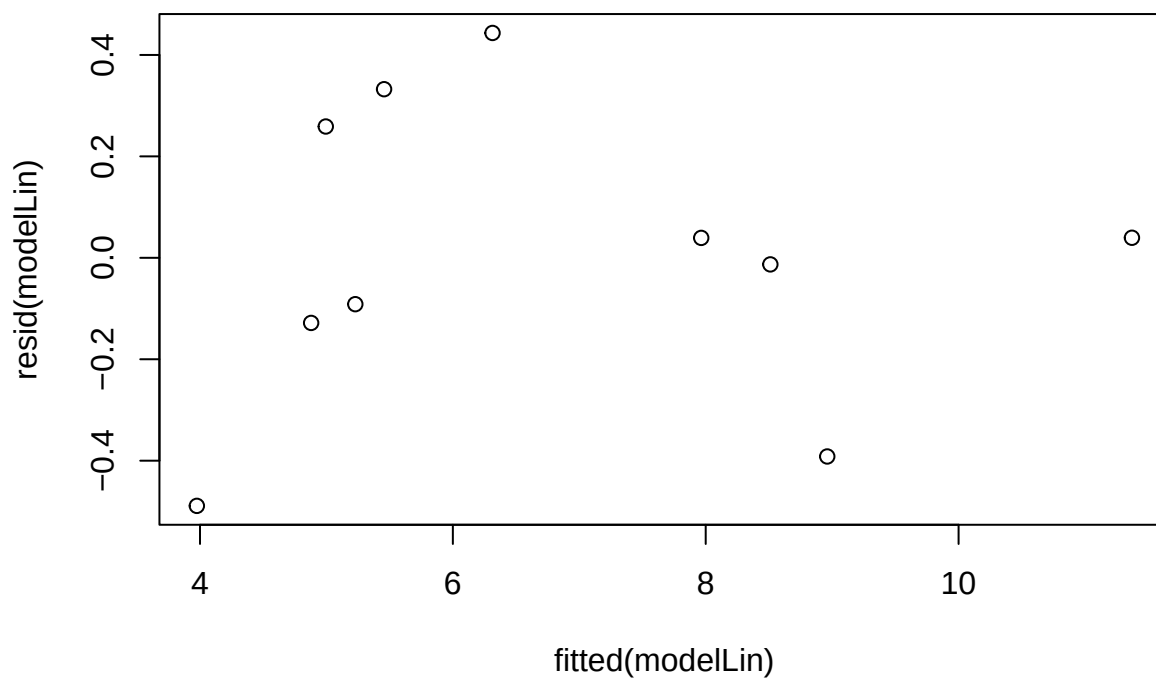
```
standard_error <- sqrt(deviance(modellin)/df.residual(modellin))
standard_error
```

```
## [1] 0.4457633
```

```
2*standard_error
```

```
## [1] 0.8915266
```

```
plot(fitted(modellin), resid(modellin))
abline(h=2*standard_error, col = "blue")
abline(h=-2*standard_error, col = "blue")
abline(h=3*standard_error, col = "red")
abline(h=-3*standard_error, col = "red")
```



```
t <- qt(df = 16 - 5 - 2, 0.95)
t
```

```
## [1] 1.833113
```

```
jackknife <- rstudent(modellin)
sort(round(jackknife, 4))
```

```
##      1      8      9      4      5      7      2      3      6      10
## -5.2063 -1.8840 -0.6878 -0.3431 -0.2592  0.1047  0.6938  0.9450  1.6289  3.9584
```

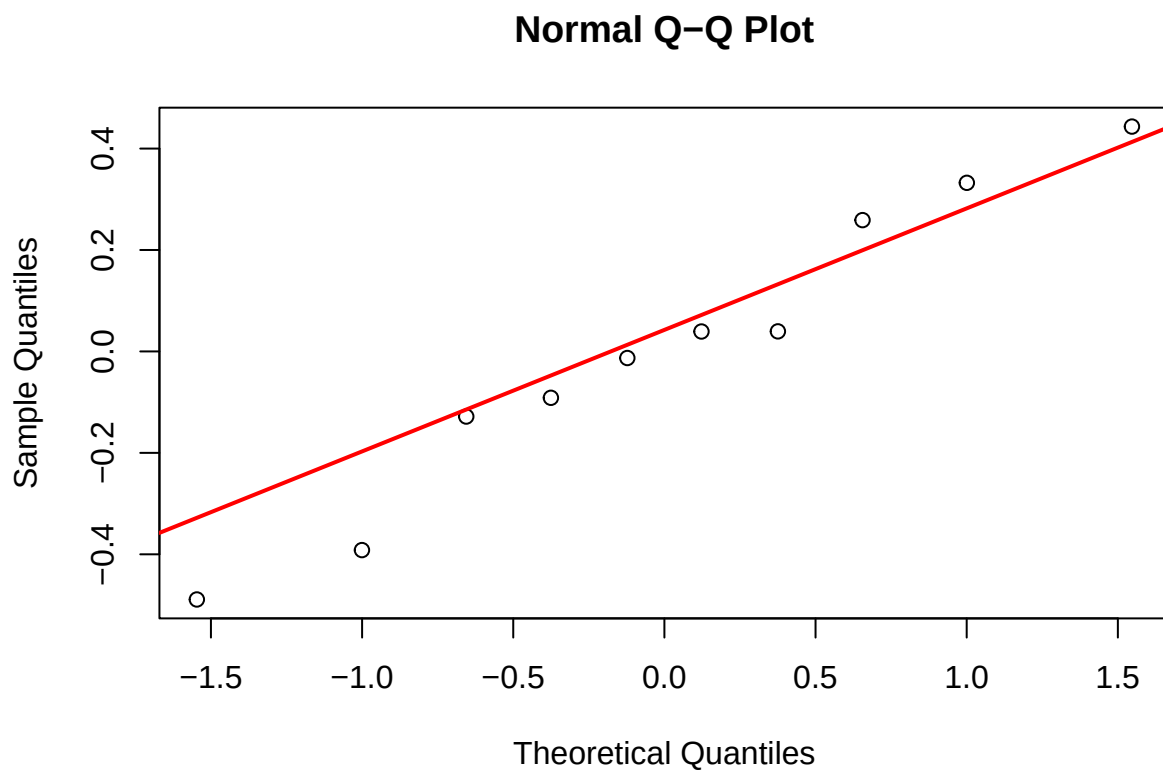
```
cookCV <- 4/16
cookCV
```

```
## [1] 0.25
```

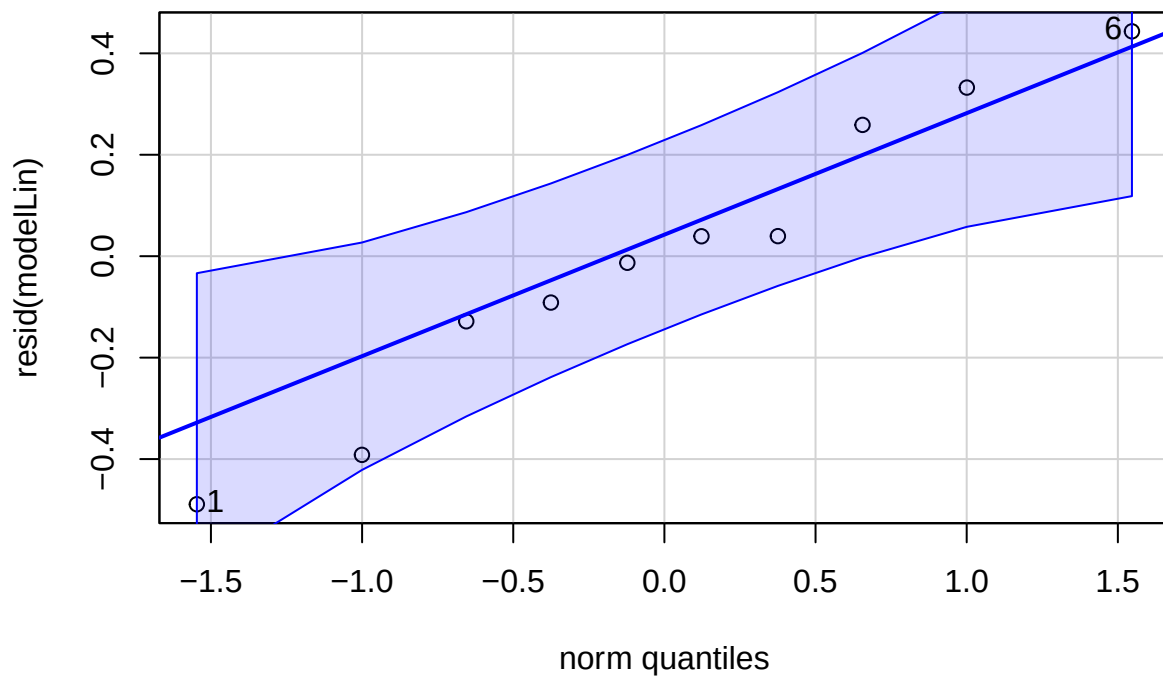
```
cook <- cooks.distance(modellin)
sort(round(cook, 4))
```

```
##      7      5      4      2      3      6      8      1
##  0.0021  0.0158  0.0207  0.0590  0.0993  0.2810  0.6532  1.1956
##      9      10
## 58.4443 238.1892
```

```
qqnorm(resid(modellin))
qqline(resid(modellin), col = "red", lwd = 2)
```



```
qqPlot(resid(modellin))
```



```
## [1] 1 6
```

```
skewness(jackknife)
```

```
## [1] -0.5625334
```

```
kurtosis(jackknife)
```

```
## [1] 3.744878
```

```
ols_vif_tol(modellin)
```

```
##      Variables  Tolerance      VIF
## 1      USAge 0.04830445 20.702027
## 2      USCar 0.13006525  7.688449
## 3      USGas 0.38925744  2.568994
## 4 PrivateMiles 0.16797585  5.953237
## 5 PublicMiles 0.02915573 34.298570
```

```
eigprop(modellin)
```

```
##
```

```
## Call:
## eigprop(mod = modelLin)
##
##      Eigenvalues      CI (Intercept)  USAge  USCar  USGas  PrivateMiles
## 1      5.9199      1.0000      0.0000 0.0000 0.0000 0.0004      0.0000
## 2      0.0508     10.7991      0.0000 0.0000 0.0030 0.0034      0.0000
## 3      0.0274     14.6959      0.0000 0.0000 0.0001 0.3645      0.0011
## 4      0.0012     71.4066      0.0005 0.0008 0.4941 0.4394      0.0057
## 5      0.0007     90.6478      0.0017 0.0008 0.0126 0.1894      0.2557
## 6      0.0000    1384.5294      0.9978 0.9984 0.4902 0.0029      0.7375
##      PublicMiles
## 1      0.0000
## 2      0.0230
## 3      0.0000
## 4      0.0711
## 5      0.0132
## 6      0.8927
##
## =====
## Row 6==> USAge, proportion 0.998389 >= 0.50
## Row 6==> PrivateMiles, proportion 0.737494 >= 0.50
## Row 6==> PublicMiles, proportion 0.892684 >= 0.50
```

```
ols_step_all_possible(modelLin)
```

##	Index	N	Predictors	R-Square	Adj. R-Square
## 1	1	1	USAge	0.90795539	0.89644982
## 5	2	1	PublicMiles	0.61395163	0.56569559
## 2	3	1	USCar	0.20652022	0.10733525
## 4	4	1	PrivateMiles	0.09919021	-0.01341102
## 3	5	1	USGas	0.06949315	-0.04682020
## 7	6	2	USAge USGas	0.96687996	0.95741709
## 6	7	2	USAge USCar	0.92581176	0.90461511
## 8	8	2	USAge PrivateMiles	0.91097511	0.88553943
## 9	9	2	USAge PublicMiles	0.90857921	0.88245898
## 15	10	2	PrivateMiles PublicMiles	0.86937555	0.83205428
## 14	11	2	USGas PublicMiles	0.63469300	0.53031957
## 12	12	2	USCar PublicMiles	0.63367250	0.52900750
## 10	13	2	USCar USGas	0.41781841	0.25148081
## 11	14	2	USCar PrivateMiles	0.38416349	0.20821020
## 13	15	2	USGas PrivateMiles	0.12347243	-0.12696402
## 19	16	3	USAge USGas PrivateMiles	0.98407836	0.97611755
## 20	17	3	USAge USGas PublicMiles	0.97786150	0.96679224
## 16	18	3	USAge USCar USGas	0.96804398	0.95206597
## 18	19	3	USAge USCar PublicMiles	0.94096724	0.91145086
## 17	20	3	USAge USCar PrivateMiles	0.93726389	0.90589584
## 21	21	3	USAge PrivateMiles PublicMiles	0.92335904	0.88503857
## 24	22	3	USCar PrivateMiles PublicMiles	0.87972795	0.81959193
## 25	23	3	USGas PrivateMiles PublicMiles	0.87601353	0.81402030
## 23	24	3	USCar USGas PublicMiles	0.63811863	0.45717795
## 22	25	3	USCar USGas PrivateMiles	0.49915254	0.24872882
## 26	26	4	USAge USCar USGas PrivateMiles	0.98412577	0.97142639
## 29	27	4	USAge USGas PrivateMiles PublicMiles	0.98410494	0.97138890
## 27	28	4	USAge USCar USGas PublicMiles	0.98064609	0.96516296

```
## 28      29 4      USAge USCar PrivateMiles PublicMiles 0.94097497      0.89375495
## 30      30 4      USCar USGas PrivateMiles PublicMiles 0.92810125      0.87058224
## 31      31 5 USAge USCar USGas PrivateMiles PublicMiles 0.98412770      0.96428732
##      Mallow's Cp
## 1      0.83978082
## 5      0.44460758
## 2     -4.58254356
## 4     -0.42286745
## 3     -0.40300929
## 7      0.93268854
## 6      0.57385891
## 8      0.71324656
## 9      0.75630312
## 15     0.73890428
## 14     0.20820842
## 12    -2.91317028
## 10    -4.34947294
## 11    -9.86316095
## 13    -2.06488855
## 19     0.96334872
## 20     0.90823360
## 16     0.52278880
## 18     0.75287077
## 17    -1.13338525
## 21     0.27366583
## 24     0.77581647
## 25    -0.09351491
## 23    -4.04768791
## 22   -12.31735191
## 26     0.66807929
## 29     0.93511951
## 27     0.65086416
## 28   -16.63449336
## 30    -4.42970983
## 31    -6.17044877
```

```
ols_step_forward_p(modellin)
```

```
##
##
##                               Stepwise Summary
## -----
## Step      Variable           AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0          Base Model         48.488   49.093   16.344   0.00000   0.00000
## 1          USAge              26.633   27.541   -4.668   0.90796   0.89645
## 2          USGas              18.412   19.622   -8.857   0.96688   0.95742
## 3          PrivateMiles       13.088   14.600   -7.807   0.98408   0.97612
## -----
##
## Final Model Output
## -----
##
##                               Model Summary
```

```
## -----
## R                0.992      RMSE                0.282
## R-Squared        0.984      MSE                  0.133
## Adj. R-Squared   0.976      Coef. Var            5.388
## Pred R-Squared   0.963      AIC                  13.088
## MAE              0.224      SBC                  14.600
## -----
```

```
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
##
```

ANOVA

```
## -----
##              Sum of
##              Squares      DF      Mean Square      F      Sig.
## -----
## Regression    49.279        3          16.426    123.615    0.0000
## Residual       0.797        6           0.133
## Total         50.076        9
## -----
```

Parameter Estimates

```
## -----
##      model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept)  -218.376      12.321              -17.724    0.000      -248.524    -188.228
##      USAge      5.606       0.311              18.009    0.000       4.844      6.368
##      USGas      1.404       0.268              5.249    0.002       0.750      2.059
## PrivateMiles  0.002       0.001              2.546    0.044       0.000      0.003
## -----
```

```
ols_step_backward_p(modellin)
```

```
##
##
##              Stepwise Summary
## -----
## Step  Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0      Full Model    17.057    19.175    2.178    0.98413    0.96429
## 1      PublicMiles   15.058    16.873   -2.822    0.98413    0.97143
## 2      USCar         13.088    14.600   -7.807    0.98408    0.97612
## -----
```

```
##
## Final Model Output
## -----
```

```
##
##              Model Summary
## -----
## R                0.992      RMSE                0.282
## R-Squared        0.984      MSE                  0.133
## Adj. R-Squared   0.976      Coef. Var            5.388
```

```
## Pred R-Squared      0.963      AIC      13.088
## MAE                 0.224      SBC      14.600
```

```
## -----
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
```

```
##
## ANOVA
```

```
## -----
##          Sum of
##          Squares      DF      Mean Square      F      Sig.
## -----
## Regression      49.279      3      16.426      123.615      0.0000
## Residual         0.797      6      0.133
## Total           50.076      9
```

```
## -----
## Parameter Estimates
## -----
##          model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept)    -218.376      12.321              -17.724      0.000      -248.524      -188.228
##      USAge       5.606       0.311       1.076      18.009      0.000       4.844       6.368
##      USGas       1.404       0.268       0.326       5.249      0.002       0.750       2.059
## PrivateMiles    0.002       0.001       0.143       2.546      0.044       0.000       0.003
## -----
```

```
ols_step_both_p(modellin)
```

```
##
##
## Stepwise Summary
## -----
## Step      Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0      Base Model      48.488      49.093      16.344      0.00000      0.00000
## 1      USAge (+)      26.633      27.541      -4.668      0.90796      0.89645
## 2      USGas (+)      18.412      19.622      -8.857      0.96688      0.95742
## 3      PrivateMiles (+) 13.088      14.600      -7.807      0.98408      0.97612
## -----
```

```
##
## Final Model Output
## -----
```

```
##
## Model Summary
## -----
## R      0.992      RMSE      0.282
## R-Squared      0.984      MSE      0.133
## Adj. R-Squared      0.976      Coef. Var      5.388
## Pred R-Squared      0.963      AIC      13.088
## MAE      0.224      SBC      14.600
## -----
```

```
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
```

```
##
```

```
## ANOVA
```

```
## -----
##          Sum of
##          Squares      DF      Mean Square      F      Sig.
## -----
## Regression      49.279      3          16.426     123.615     0.0000
## Residual         0.797      6           0.133
## Total           50.076      9
```

```
## -----
```

```
##
```

```
## Parameter Estimates
```

```
## -----
##          model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept)    -218.376      12.321              -17.724     0.000     -248.524     -188.228
##      USAge         5.606       0.311              18.009     0.000       4.844       6.368
##      USGas         1.404       0.268              5.249     0.002       0.750       2.059
## PrivateMiles     0.002       0.001              2.546     0.044       0.000       0.003
```

```
## -----
```

```
ols_step_best_subset(modellin)
```

```
## Best Subsets Regression
```

```
## -----
## Model Index      Predictors
## -----
##      1      USAge
##      2      USAge USGas
##      3      USAge USGas PrivateMiles
##      4      USAge USCar USGas PrivateMiles
##      5      USAge USCar USGas PrivateMiles PublicMiles
```

```
## -----
```

```
##
```

```
## Subsets Regression Summary
```

```
## -----
##          Model      R-Square      Adj.      Pred
##          Model      R-Square      R-Square      R-Square      C(p)      AIC      SBIC      SBC      MSEP      I
## -----
##      1      0.9080      0.8964      0.8398      17.1963      26.6335      -4.6681      27.5413      5.7944      0
##      2      0.9669      0.9574      0.9327      4.3466      18.4121      -8.8566      19.6225      2.4325      0
##      3      0.9841      0.9761      0.9633      2.0124      13.0876      -7.8068      14.6005      1.4032      0
##      4      0.9841      0.9714      0.6681      4.0005      15.0577      -2.8223      16.8732      1.7488      0
##      5      0.9841      0.9643      -6.1704      6.0000      17.0565      2.1777      19.1746      2.3315      0
```

```
## -----
```

```
## AIC: Akaike Information Criteria
```

```
## SBIC: Sawa's Bayesian Information Criteria
```

```
## SBC: Schwarz Bayesian Criteria
```

```
## MSEP: Estimated error of prediction, assuming multivariate normality
```

```
## FPE: Final Prediction Error
## HSP: Hocking's Sp
## APC: Amemiya Prediction Criteria
```

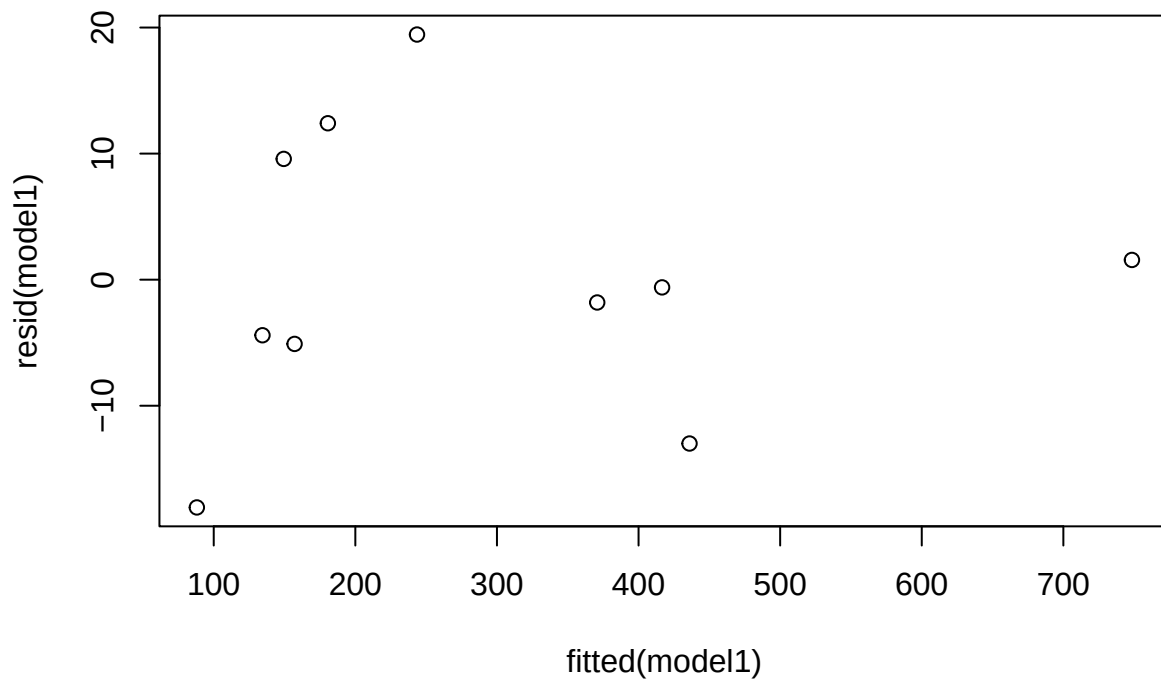
```
standard_error <- sqrt(deviance(model1)/df.residual(model1))
standard_error
```

```
## [1] 17.1075
```

```
2*standard_error
```

```
## [1] 34.21499
```

```
plot(fitted(model1),resid(model1))
abline(h=2*standard_error, col = "blue")
abline(h=-2*standard_error, col = "blue")
abline(h=3*standard_error, col = "red")
abline(h=-3*standard_error, col = "red")
```



```
t <- qt(df = 16 - 5 - 2, 0.95)
t
```

```
## [1] 1.833113
```

```
jackknife <- rstudent(model1)
sort(round(jackknife, 4))
```

```
##      1      8      9      5      4      7      2      3      6      10
## -3.8859 -1.4295 -0.8851 -0.3813 -0.3059 -0.1258  0.6652  0.9116  2.1798  4.9587
```

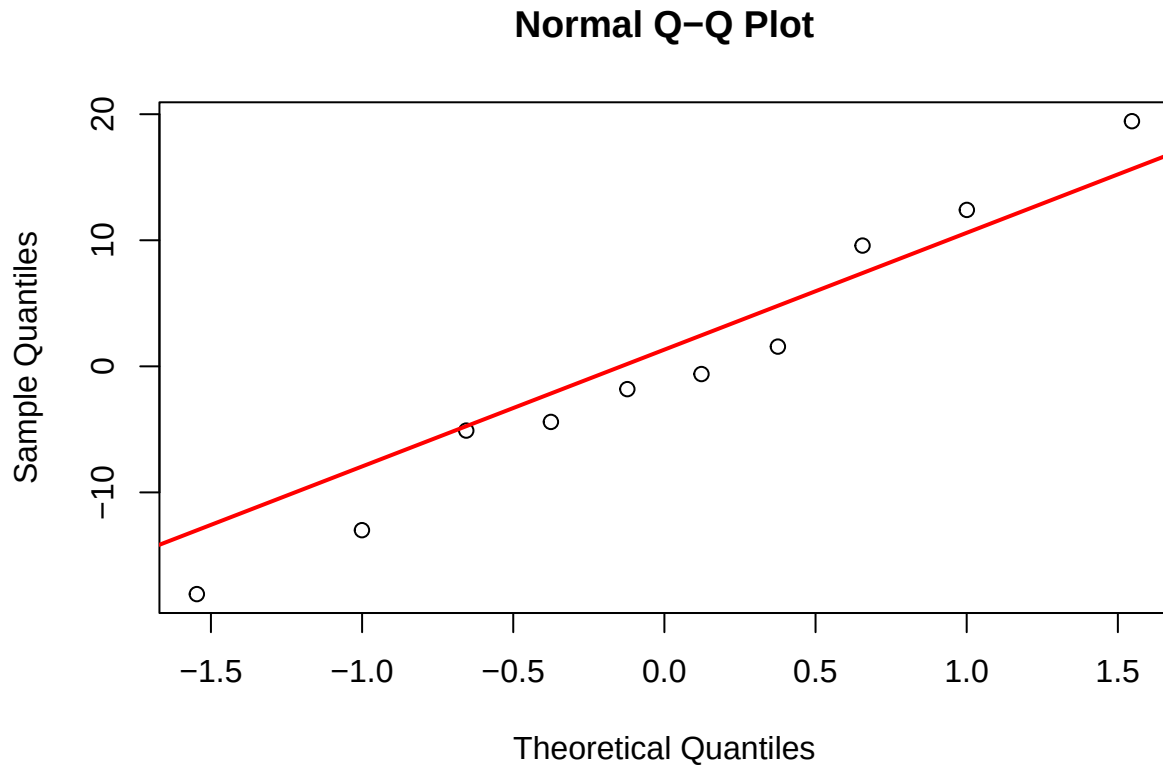
```
cookCV <- 4/16
cookCV
```

```
## [1] 0.25
```

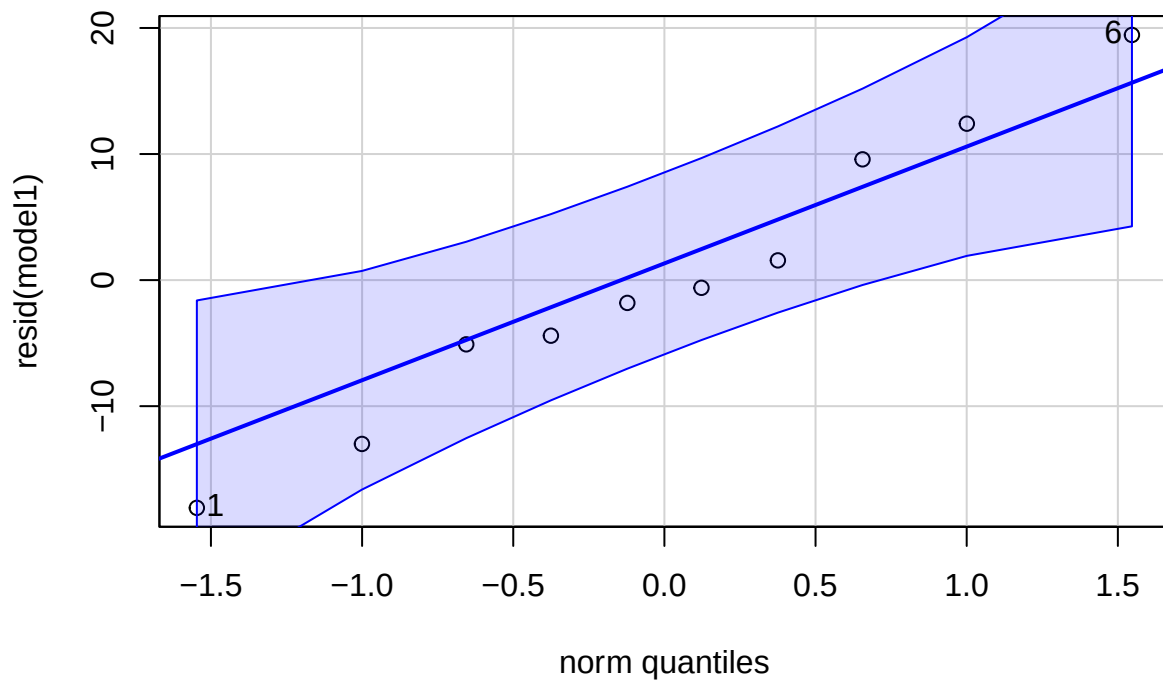
```
cook <- cooks.distance(model1)
sort(round(cook, 4))
```

```
##      7      4      5      2      3      6      8      1
##  0.0031  0.0166  0.0334  0.0548  0.0939  0.3669  0.4883  1.1079
##      9      10
## 88.8501 252.9326
```

```
qqnorm(resid(model1))
qqline(resid(model1), col = "red", lwd = 2)
```



```
qqPlot(resid(model1))
```



```
## [1] 6 1
```

```
skewness(jackknife)
```

```
## [1] 0.4319298
```

```
kurtosis(jackknife)
```

```
## [1] 3.489636
```

```
ols_vif_tol(model1)
```

```
##      Variables Tolerance      VIF
## 1      USAge 0.04830445 20.702027
## 2      USCar 0.13006525  7.688449
## 3      USGas 0.38925744  2.568994
## 4 PrivateMiles 0.16797585  5.953237
## 5 PublicMiles 0.02915573 34.298570
```

```
eigprop(model1)
```

```
##
## Call:
## eigprop(mod = model1)
##
##   Eigenvalues      CI (Intercept)  USAge  USCar  USGas PrivateMiles
## 1      5.9199      1.0000      0.0000 0.0000 0.0000 0.0004      0.0000
## 2      0.0508     10.7991      0.0000 0.0000 0.0030 0.0034      0.0000
## 3      0.0274     14.6959      0.0000 0.0000 0.0001 0.3645      0.0011
## 4      0.0012     71.4066      0.0005 0.0008 0.4941 0.4394      0.0057
## 5      0.0007     90.6478      0.0017 0.0008 0.0126 0.1894      0.2557
## 6      0.0000    1384.5294      0.9978 0.9984 0.4902 0.0029      0.7375
##   PublicMiles
## 1      0.0000
## 2      0.0230
## 3      0.0000
## 4      0.0711
## 5      0.0132
## 6      0.8927
##
## =====
## Row 6==> USAge, proportion 0.998389 >= 0.50
## Row 6==> PrivateMiles, proportion 0.737494 >= 0.50
## Row 6==> PublicMiles, proportion 0.892684 >= 0.50
```

```
ols_step_all_possible(model1)
```

```
##      Index N      Predictors  R-Square Adj. R-Square
## 1      1 1      USAge 0.86707844 0.85046324
## 5      2 1      PublicMiles 0.70234165 0.66513436
## 2      3 1      USCar 0.33674666 0.25384000
## 4      4 1      PrivateMiles 0.06980714 -0.04646697
## 3      5 1      USGas 0.03338182 -0.08744545
## 7      6 2      USAge USGas 0.97177889 0.96371571
## 6      7 2      USAge USCar 0.94309643 0.92683827
## 15     8 2      PrivateMiles PublicMiles 0.91947336 0.89646574
## 9      9 2      USAge PublicMiles 0.88864865 0.85683398
## 8     10 2      USAge PrivateMiles 0.86714882 0.82919134
## 14     11 2      USGas PublicMiles 0.70521231 0.62098725
## 12     12 2      USCar PublicMiles 0.70263021 0.61766741
## 10     13 2      USCar USGas 0.51467904 0.37601591
## 11     14 2      USCar PrivateMiles 0.49435761 0.34988835
## 13     15 2      USGas PrivateMiles 0.07776016 -0.18573694
## 19     16 3      USAge USGas PrivateMiles 0.98323352 0.97485027
## 16     17 3      USAge USCar USGas 0.97881740 0.96822609
## 20     18 3      USAge USGas PublicMiles 0.97188126 0.95782189
## 17     19 3      USAge USCar PrivateMiles 0.95499832 0.93249749
## 18     20 3      USAge USCar PublicMiles 0.95390308 0.93085462
## 25     21 3      USGas PrivateMiles PublicMiles 0.94600072 0.91900108
## 21     22 3      USAge PrivateMiles PublicMiles 0.92961026 0.89441540
## 24     23 3      USCar PrivateMiles PublicMiles 0.91983497 0.87975245
```

```

## 23      24 3              USCar USGas PublicMiles 0.70589006      0.55883509
## 22      25 3              USCar USGas PrivateMiles 0.58896382      0.38344574
## 29      26 4      USAge USGas PrivateMiles PublicMiles 0.99644102      0.99359384
## 26      27 4              USAge USCar USGas PrivateMiles 0.99503595      0.99106471
## 27      28 4              USAge USCar USGas PublicMiles 0.98764672      0.97776410
## 30      29 4      USCar USGas PrivateMiles PublicMiles 0.96525498      0.93745896
## 28      30 4      USAge USCar PrivateMiles PublicMiles 0.95527850      0.91950129
## 31      31 5 USAge USCar USGas PrivateMiles PublicMiles 0.99685039      0.99291339
##      Mallow's Cp
## 1      0.7167970
## 5      0.4573554
## 2      -3.8180725
## 4      -0.3700261
## 3      -0.4063592
## 7      0.9115798
## 6      0.5632423
## 15     0.8693891
## 9      0.6126794
## 8      0.3961999
## 14     0.2313448
## 12     -2.4335653
## 10     -3.6624464
## 11     -8.2382806
## 13     -1.7992424
## 19     0.8539886
## 16     0.5141424
## 20     0.8379129
## 17     -0.5242939
## 18     0.6196814
## 25     0.5484357
## 21     0.1392574
## 24     0.8310066
## 23     -3.4434430
## 22    -10.4727241
## 29     0.9591187
## 26     0.5925111
## 27     0.5116725
## 30     -2.9174470
## 28     -9.8632591
## 31     -0.6355876

```

```
ols_step_forward_p(model1)
```

```

##
##
##      Stepwise Summary
## -----
## Step   Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0      Base Model    137.611  138.216  105.279  0.00000  0.00000
## 1      USAge         119.431  120.339   85.519  0.86708  0.85046
## 2      USGas         105.934  107.144   72.190  0.97178  0.96372
## 3      PrivateMiles   102.727  104.240   69.543  0.98323  0.97485
## 4      PublicMiles    89.228   91.044   70.034  0.99644  0.99359

```

```
## -----
##
## Final Model Output
## -----
##
##                               Model Summary
## -----
## R                               0.998      RMSE                11.501
## R-Squared                       0.996      MSE                 264.565
## Adj. R-Squared                   0.994      Coef. Var            5.561
## Pred R-Squared                   0.959      AIC                   89.228
## MAE                              9.277      SBC                   91.044
## -----
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
##
##                               ANOVA
## -----
##                               Sum of
##                               Squares      DF      Mean Square      F      Sig.
## -----
## Regression    370363.676      4      92590.919    349.975    0.0000
## Residual      1322.824      5       264.565
## Total         371686.500      9
## -----
##
##                               Parameter Estimates
## -----
##                               model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept)   -13808.769      1439.403              -9.593      0.000      -17508.873      -10108.664
## USAge         334.847      39.777       0.746      8.418      0.000      232.596      437.098
## USGas         128.059      13.216       0.345      9.690      0.000      94.086      162.032
## PrivateMiles   0.245      0.042       0.269      5.874      0.002      0.138      0.352
## PublicMiles   -5.822      1.352      -0.351     -4.308      0.008      -9.296      -2.348
## -----
```

```
ols_step_backward_p(model1)
```

```
##
##
##                               Stepwise Summary
## -----
## Step      Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0      Full Model      90.006      92.124      75.127      0.99685      0.99291
## 1      USCar          89.228      91.044      70.034      0.99644      0.99359
## -----
##
## Final Model Output
## -----
```

```
##
##                               Model Summary
## -----
## R                               0.998      RMSE                11.501
## R-Squared                       0.996      MSE                264.565
## Adj. R-Squared                   0.994      Coef. Var           5.561
## Pred R-Squared                   0.959      AIC                  89.228
## MAE                             9.277      SBC                  91.044
## -----
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
##
##                               ANOVA
## -----
##                               Sum of
##                               Squares      DF      Mean Square      F      Sig.
## -----
## Regression      370363.676           4      92590.919      349.975      0.0000
## Residual         1322.824           5           264.565
## Total           371686.500           9
## -----
##
##                               Parameter Estimates
## -----
##                               model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept)    -13808.769      1439.403              -9.593      0.000      -17508.873      -10108.664
## USAge           334.847        39.777           0.746      8.418      0.000           232.596           437.098
## USGas           128.059        13.216           0.345      9.690      0.000           94.086           162.032
## PrivateMiles     0.245         0.042           0.269      5.874      0.002            0.138            0.352
## PublicMiles     -5.822         1.352          -0.351     -4.308      0.008           -9.296           -2.348
## -----
```

```
ols_step_both_p(model1)
```

```
##
##
##                               Stepwise Summary
## -----
## Step      Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0      Base Model      137.611      138.216      105.279      0.00000      0.00000
## 1      USAge (+)      119.431      120.339      85.519      0.86708      0.85046
## 2      USGas (+)      105.934      107.144      72.190      0.97178      0.96372
## 3      PrivateMiles (+) 102.727      104.240      69.543      0.98323      0.97485
## 4      PublicMiles (+) 89.228       91.044      70.034      0.99644      0.99359
## -----
##
## Final Model Output
## -----
##
```

```
##                               Model Summary
## -----
## R                               0.998      RMSE                11.501
## R-Squared                       0.996      MSE                264.565
## Adj. R-Squared                  0.994      Coef. Var           5.561
## Pred R-Squared                  0.959      AIC                 89.228
## MAE                             9.277      SBC                 91.044
## -----
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
##
##                               ANOVA
## -----
##                               Sum of
##                               Squares      DF      Mean Square      F      Sig.
## -----
## Regression    370363.676      4      92590.919    349.975    0.0000
## Residual      1322.824      5       264.565
## Total        371686.500      9
## -----
##
##                               Parameter Estimates
## -----
##                               model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept)   -13808.769      1439.403              -9.593      0.000      -17508.873      -10108.664
## USAge          334.847       39.777       0.746      8.418      0.000       232.596       437.098
## USGas          128.059       13.216       0.345      9.690      0.000       94.086       162.032
## PrivateMiles    0.245        0.042       0.269      5.874      0.002        0.138        0.352
## PublicMiles    -5.822        1.352      -0.351     -4.308      0.008       -9.296       -2.348
## -----
```

```
ols_step_best_subset(model1)
```

```
##                               Best Subsets Regression
## -----
## Model Index    Predictors
## -----
##      1         USAge
##      2         USAge USGas
##      3         USAge USGas PrivateMiles
##      4         USAge USGas PrivateMiles PublicMiles
##      5         USAge USCar USGas PrivateMiles PublicMiles
## -----
##
##                               Subsets Regression Summary
## -----
##                               Adj.      Pred
##                               R-Square  R-Square      C(p)      AIC      SBIC      SBC      MSEP
## -----
##      1         0.8671      0.8505      0.7168    162.8104    119.4310    85.5191    120.3388    62109.33
```

```
##      2      0.9718      0.9637      0.9116      31.8408      105.9341      72.1898      107.1445      15384.46
##      3      0.9832      0.9749      0.854      19.2934      102.7272      69.5429      104.2402      10968.10
##      4      0.9964      0.9936      0.9591      4.5199      89.2282      70.0338      91.0437      2910.21
##      5      0.9969      0.9929      -0.6356      6.0000      90.0062      75.1274      92.1243      3433.95
## -----
## AIC: Akaike Information Criteria
## SBIC: Sawa's Bayesian Information Criteria
## SBC: Schwarz Bayesian Criteria
## MSEP: Estimated error of prediction, assuming multivariate normality
## FPE: Final Prediction Error
## HSP: Hocking's Sp
## APC: Amemiya Prediction Criteria
```

```
model2 <- lm(US ~ USAge + USGas + PrivateMiles + PublicMiles, data = ebikes)
model2
```

```
##
## Call:
## lm(formula = US ~ USAge + USGas + PrivateMiles + PublicMiles,
##     data = ebikes)
##
## Coefficients:
## (Intercept)      USAge      USGas PrivateMiles PublicMiles
## -13808.769      334.847      128.059         0.245        -5.822
```

```
summary(model2)
```

```
##
## Call:
## lm(formula = US ~ USAge + USGas + PrivateMiles + PublicMiles,
##     data = ebikes)
##
## Residuals:
##      1      2      3      4      5      6      7      8
## -23.0272  11.5941  16.0890   0.4507  -1.8890  13.9472  -6.9400 -11.0545
##      9     10
##  -3.4765   4.3062
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.381e+04  1.439e+03  -9.593 0.000209 ***
## USAge        3.348e+02  3.978e+01   8.418 0.000388 ***
## USGas        1.281e+02  1.322e+01   9.690 0.000199 ***
## PrivateMiles 2.450e-01  4.170e-02   5.874 0.002030 **
## PublicMiles  -5.822e+00  1.352e+00  -4.308 0.007660 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.27 on 5 degrees of freedom
## (6 observations deleted due to missingness)
## Multiple R-squared:  0.9964, Adjusted R-squared:  0.9936
## F-statistic: 350 on 4 and 5 DF, p-value: 2.638e-06
```

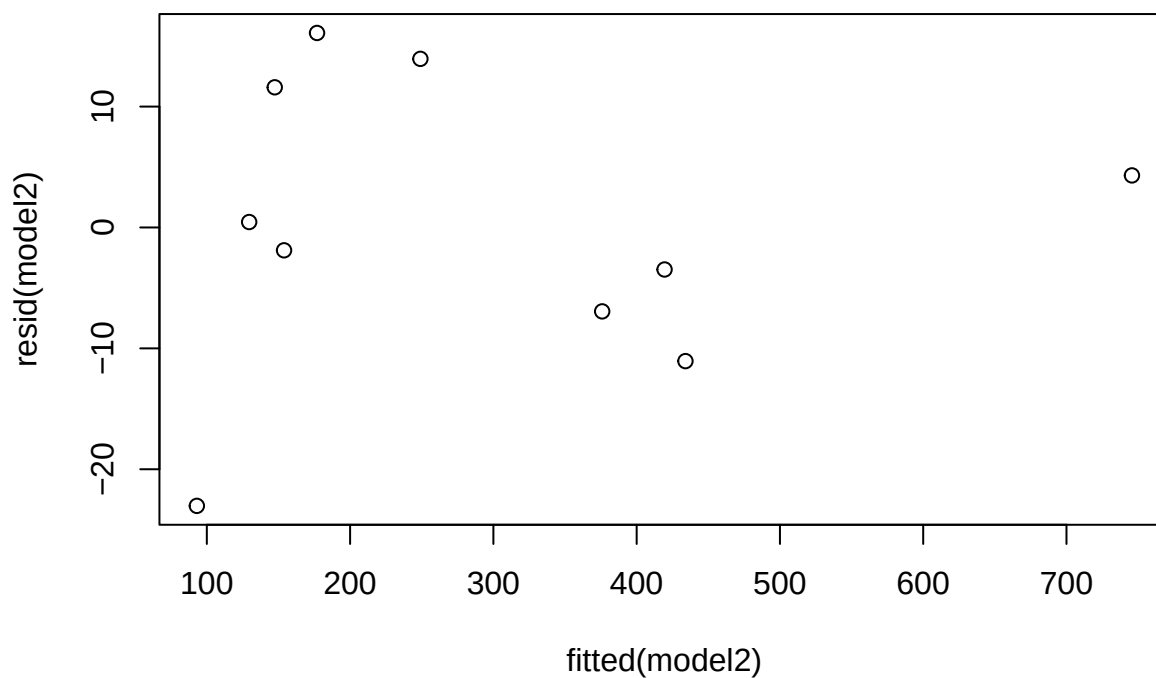
```
standard_error <- sqrt(deviance(model2)/df.residual(model2))
standard_error
```

```
## [1] 16.26545
```

```
2*standard_error
```

```
## [1] 32.53089
```

```
plot(fitted(model2), resid(model2))
abline(h=2*standard_error, col = "blue")
abline(h=-2*standard_error, col = "blue")
abline(h=3*standard_error, col = "red")
abline(h=-3*standard_error, col = "red")
```



```
t <- qt(df = 16 - 4 - 2, 0.95)
t
```

```
## [1] 1.812461
```

```
jackknife <- rstudent(model2)
sort(round(jackknife, 4))
```

```
##      1      8      9      7      5      4      2      6      10      3
## -4.1067 -1.1316 -0.8885 -0.4662 -0.1407  0.0295  0.8717  1.0084  1.2187  1.2517
```

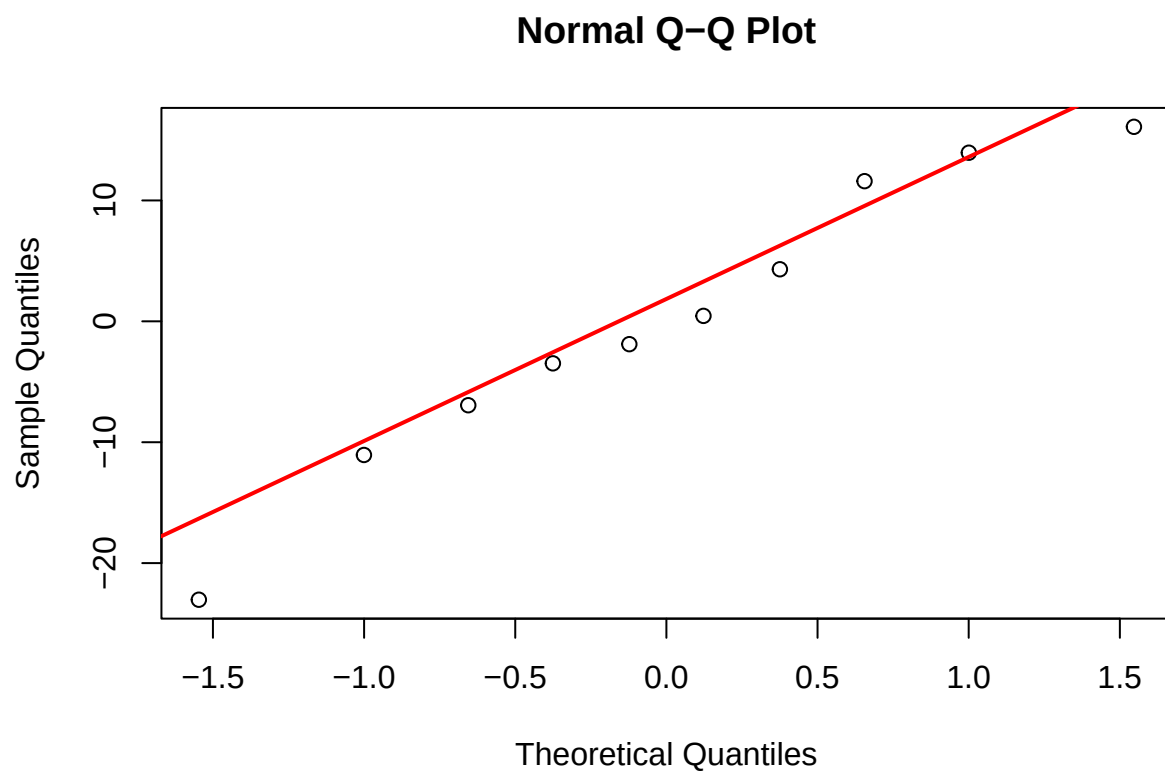
```
cookCV <- 4/16
cookCV
```

```
## [1] 0.25
```

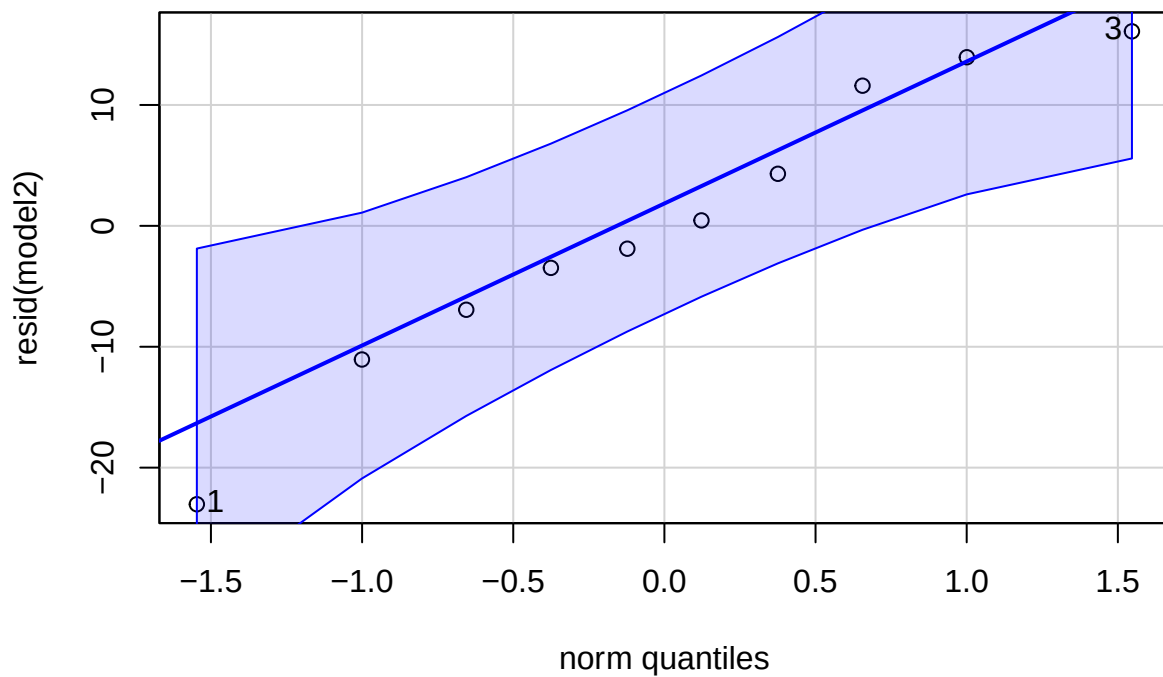
```
cook <- cooks.distance(model2)
sort(round(cook, 4))
```

```
##      4      5      7      6      2      3      8      1      9      10
## 0.0001 0.0041 0.0214 0.0767 0.0912 0.1233 0.3940 0.8216 2.8087 4.9589
```

```
qqnorm(resid(model2))
qqline(resid(model2), col = "red", lwd = 2)
```



```
qqPlot(resid(model2))
```



```
## [1] 1 3
```

```
jackknife <- rstudent(model2)
sort(round(jackknife, 4))
```

```
##      1      8      9      7      5      4      2      6      10      3
## -4.1067 -1.1316 -0.8885 -0.4662 -0.1407  0.0295  0.8717  1.0084  1.2187  1.2517
```

```
skewness(jackknife)
```

```
## [1] -1.375026
```

```
kurtosis(jackknife)
```

```
## [1] 4.341869
```

```
ols_vif_tol(model2)
```

```
##      Variables  Tolerance      VIF
## 1      USAge 0.09067784 11.028052
## 2      USGas 0.56218404  1.778777
## 3 PrivateMiles 0.33904625  2.949450
## 4 PublicMiles 0.10744047  9.307480
```

```
eigprop(model2)
```

```
##
## Call:
## eigprop(mod = model2)
##
##      Eigenvalues      CI (Intercept)  USAge  USGas PrivateMiles PublicMiles
## 1      4.9313      1.0000      0.0000 0.0000 0.0008      0.0000      0.0002
## 2      0.0407     11.0136      0.0000 0.0000 0.0333      0.0003      0.1089
## 3      0.0273     13.4468      0.0000 0.0000 0.4961      0.0025      0.0006
## 4      0.0007     82.3433      0.0039 0.0021 0.1528      0.4803      0.0193
## 5      0.0000    903.4063      0.9961 0.9979 0.3170      0.5169      0.8710
##
## =====
## Row 5==> USAge, proportion 0.997888 >= 0.50
## Row 5==> PrivateMiles, proportion 0.516883 >= 0.50
## Row 5==> PublicMiles, proportion 0.870991 >= 0.50
```

```
ols_step_all_possible(model2)
```

```
##      Index N      Predictors  R-Square Adj. R-Square
## 1      1 1      USAge 0.86707844 0.85046324
## 4      2 1      PublicMiles 0.70234165 0.66513436
## 3      3 1      PrivateMiles 0.06980714 -0.04646697
## 2      4 1      USGas 0.03338182 -0.08744545
## 5      5 2      USAge USGas 0.97177889 0.96371571
## 10     6 2      PrivateMiles PublicMiles 0.91947336 0.89646574
## 7      7 2      USAge PublicMiles 0.88864865 0.85683398
## 6      8 2      USAge PrivateMiles 0.86714882 0.82919134
## 9      9 2      USGas PublicMiles 0.70521231 0.62098725
## 8     10 2      USGas PrivateMiles 0.07776016 -0.18573694
## 11     11 3      USAge USGas PrivateMiles 0.98323352 0.97485027
## 12     12 3      USAge USGas PublicMiles 0.97188126 0.95782189
## 14     13 3      USGas PrivateMiles PublicMiles 0.94600072 0.91900108
## 13     14 3      USAge PrivateMiles PublicMiles 0.92961026 0.89441540
## 15     15 4 USAge USGas PrivateMiles PublicMiles 0.99644102 0.99359384
##      Mallow's Cp
## 1      0.7167970
## 4      0.4573554
## 3     -0.3700261
## 2     -0.4063592
## 5      0.9115798
## 10     0.8693891
## 7      0.6126794
## 6      0.3961999
## 9      0.2313448
## 8     -1.7992424
## 11     0.8539886
## 12     0.8379129
## 14     0.5484357
## 13     0.1392574
## 15     0.9591187
```

```
ols_step_forward_p(model2)
```

```
##
##
##                               Stepwise Summary
## -----
## Step      Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0      Base Model      137.611    138.216    105.275    0.00000    0.00000
## 1      USAge      119.431    120.339     85.475    0.86708    0.85046
## 2      USGas      105.934    107.144     71.950    0.97178    0.96372
## 3      PrivateMiles    102.727    104.240     69.082    0.98323    0.97485
## 4      PublicMiles     89.228     91.044     68.849    0.99644    0.99359
## -----
##
## Final Model Output
## -----
##
##                               Model Summary
## -----
## R      0.998      RMSE      11.501
## R-Squared      0.996      MSE      264.565
## Adj. R-Squared      0.994      Coef. Var      5.561
## Pred R-Squared      0.959      AIC      89.228
## MAE      9.277      SBC      91.044
## -----
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
##
##                               ANOVA
## -----
##                               Sum of
##                               Squares      DF      Mean Square      F      Sig.
## -----
## Regression      370363.676      4      92590.919      349.975      0.0000
## Residual      1322.824      5      264.565
## Total      371686.500      9
## -----
##
##                               Parameter Estimates
## -----
## model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept)      -13808.769      1439.403      -9.593      0.000      -17508.873      -10108.664
## USAge      334.847      39.777      0.746      8.418      0.000      232.596      437.098
## USGas      128.059      13.216      0.345      9.690      0.000      94.086      162.032
## PrivateMiles      0.245      0.042      0.269      5.874      0.002      0.138      0.352
## PublicMiles      -5.822      1.352      -0.351      -4.308      0.008      -9.296      -2.348
## -----
```

```
ols_step_backward_p(model2)
```

```
## [1] "No variables have been removed from the model."
```

```
ols_step_both_p(model2)
```

```
##
```

```
##
```

```
## Stepwise Summary
```

```
## -----
```

## Step	Variable	AIC	SBC	SBIC	R2	Adj. R2
## 0	Base Model	137.611	138.216	105.275	0.00000	0.00000
## 1	USAge (+)	119.431	120.339	85.475	0.86708	0.85046
## 2	USGas (+)	105.934	107.144	71.950	0.97178	0.96372
## 3	PrivateMiles (+)	102.727	104.240	69.082	0.98323	0.97485
## 4	PublicMiles (+)	89.228	91.044	68.849	0.99644	0.99359

```
## -----
```

```
##
```

```
## Final Model Output
```

```
## -----
```

```
##
```

```
## Model Summary
```

```
## -----
```

## R	0.998	RMSE	11.501
## R-Squared	0.996	MSE	264.565
## Adj. R-Squared	0.994	Coef. Var	5.561
## Pred R-Squared	0.959	AIC	89.228
## MAE	9.277	SBC	91.044

```
## -----
```

```
## RMSE: Root Mean Square Error
```

```
## MSE: Mean Square Error
```

```
## MAE: Mean Absolute Error
```

```
## AIC: Akaike Information Criteria
```

```
## SBC: Schwarz Bayesian Criteria
```

```
##
```

```
## ANOVA
```

```
## -----
```

##	Sum of	DF	Mean Square	F	Sig.
##	Squares				
## Regression	370363.676	4	92590.919	349.975	0.0000
## Residual	1322.824	5	264.565		
## Total	371686.500	9			

```
## -----
```

```
##
```

```
## Parameter Estimates
```

```
## -----
```

##	model	Beta	Std. Error	Std. Beta	t	Sig.	lower	upper
##	(Intercept)	-13808.769	1439.403		-9.593	0.000	-17508.873	-10108.664
##	USAge	334.847	39.777	0.746	8.418	0.000	232.596	437.098

```
##          USGas      128.059      13.216      0.345      9.690      0.000      94.086      162.032
## PrivateMiles      0.245      0.042      0.269      5.874      0.002      0.138      0.352
## PublicMiles     -5.822      1.352     -0.351     -4.308      0.008     -9.296     -2.348
## -----
```

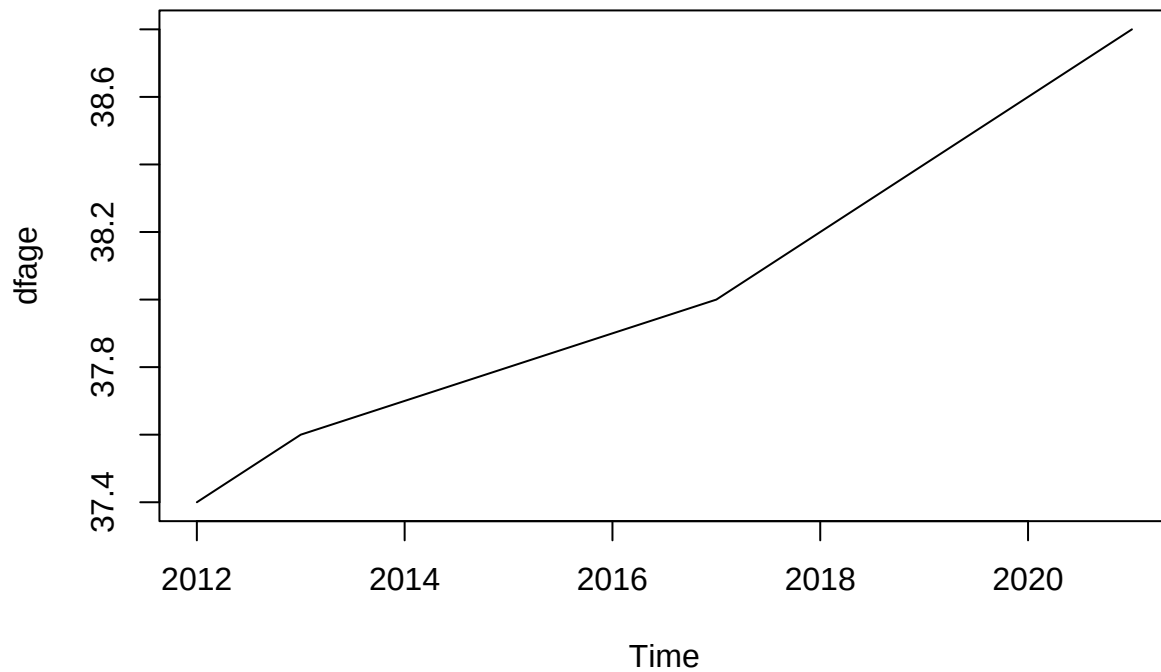
```
ols_step_best_subset(model2)
```

```
##          Best Subsets Regression
## -----
## Model Index   Predictors
## -----
##      1        USAge
##      2        USAge USGas
##      3        USAge USGas PrivateMiles
##      4        USAge USGas PrivateMiles PublicMiles
## -----
##
##                                     Subsets Regression Summary
## -----
## Model      R-Square    Adj.      Pred      C(p)      AIC      SBIC      SBC      MSEP
## -----
##      1      0.8671      0.8505      0.7168     180.7412    119.4310    85.4749    120.3388    62109.33
##      2      0.9718      0.9637      0.9116      35.6478    105.9341    71.9503    107.1445    15384.46
##      3      0.9832      0.9749      0.854      21.5552    102.7272    69.0824    104.2402    10968.10
##      4      0.9964      0.9936      0.9591      5.0000     89.2282    68.8494    91.0437    2910.21
## -----
## AIC: Akaike Information Criteria
## SBIC: Sawa's Bayesian Information Criteria
## SBC: Schwarz Bayesian Criteria
## MSEP: Estimated error of prediction, assuming multivariate normality
## FPE: Final Prediction Error
## HSP: Hocking's Sp
## APC: Amemiya Prediction Criteria
```

```
dataage <- data.frame(
  Year = c(2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021),
  USAge = c(37.4, 37.6, 37.7, 37.8, 37.9, 38.0, 38.2, 38.4, 38.6, 38.8)
)

dfage <- ts(dataage$USAge, start = min(dataage$Year), end = max(dataage$Year), frequency = 1)

plot(dfage)
```



```
model <- auto.arima(dfage, stepwise = FALSE, ic = "aic")
```

```
fit <- auto.arima(dfage)
```

```
summary(fit)
```

```
## Series: dfage
## ARIMA(0,1,0) with drift
##
## Coefficients:
##      drift
##      0.1556
## s.e.  0.0166
##
## sigma^2 = 0.002951: log likelihood = 14.25
## AIC=-24.49  AICc=-22.49  BIC=-24.1
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 0.003724443 0.04858947 0.04816889 0.009244538 0.1266681 0.3096571
##              ACF1
## Training set 0.5135724
```

```
forecast <- forecast(fit, h = 10)
```

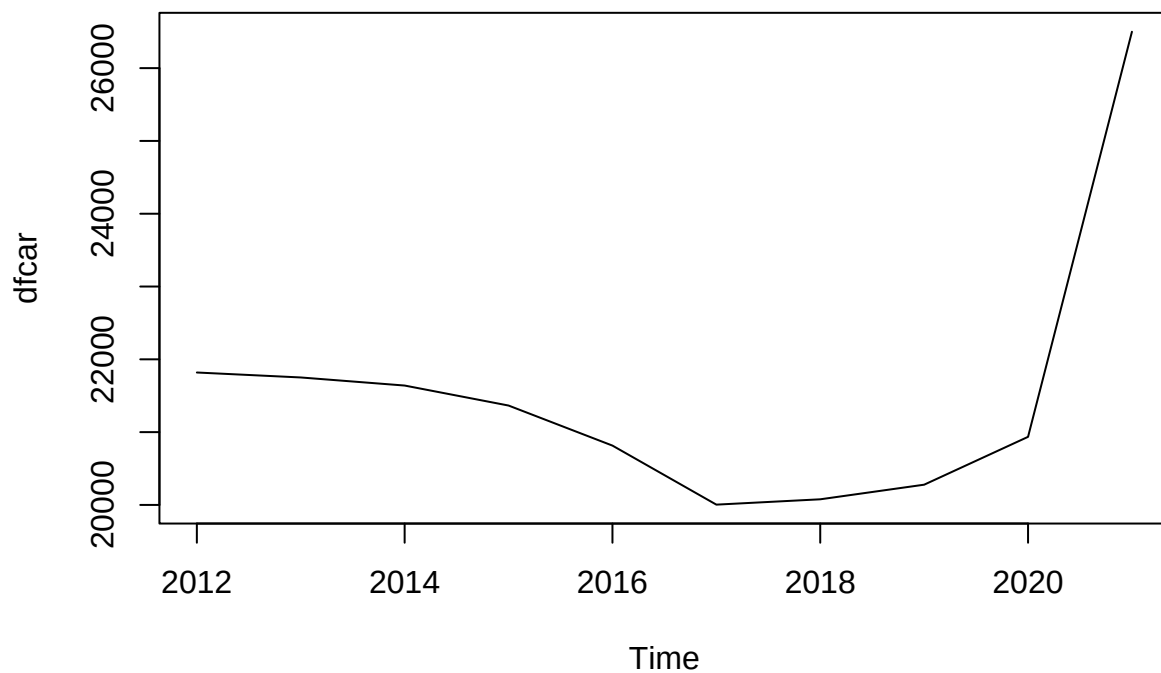
```
print(forecast)
```

```
##      Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 2022      38.95556 38.88594 39.02518 38.84908 39.06203
## 2023      39.11111 39.01265 39.20957 38.96053 39.26169
## 2024      39.26667 39.14608 39.38725 39.08225 39.45109
## 2025      39.42222 39.28298 39.56146 39.20927 39.63517
## 2026      39.57778 39.42210 39.73345 39.33969 39.81586
## 2027      39.73333 39.56280 39.90387 39.47253 39.99414
## 2028      39.88889 39.70469 40.07309 39.60718 40.17059
## 2029      40.04444 39.84753 40.24136 39.74329 40.34560
## 2030      40.20000 39.99114 40.40886 39.88058 40.51942
## 2031      40.35556 40.13540 40.57571 40.01885 40.69226
```

```
dataCar <- data.frame(
  Year = c(2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021),
  UScar = c(21819, 21751, 21640, 21365, 20815, 20004, 20077, 20278, 20935, 26500)
)
```

```
dfcar <- ts(dataCar$UScar, start = min(dataCar$Year), end = max(dataCar$Year), frequency = 1)
```

```
plot(dfcar)
```



```
model <- auto.arima(dfcar, stepwise = FALSE, ic = "aic")
```

```
fitcar <- auto.arima(dfcar)
```

```
summary(fitcar)
```

```
## Series: dfcar
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##             mean
##      21518.4000
## s.e.      563.4202
##
## sigma^2 = 3527098: log likelihood = -89.04
## AIC=182.09   AICc=183.8   BIC=182.69
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set -1.455192e-12 1781.681 1127.28 -0.5943343 4.955832 1.220734
##              ACF1
## Training set 0.09579274
```

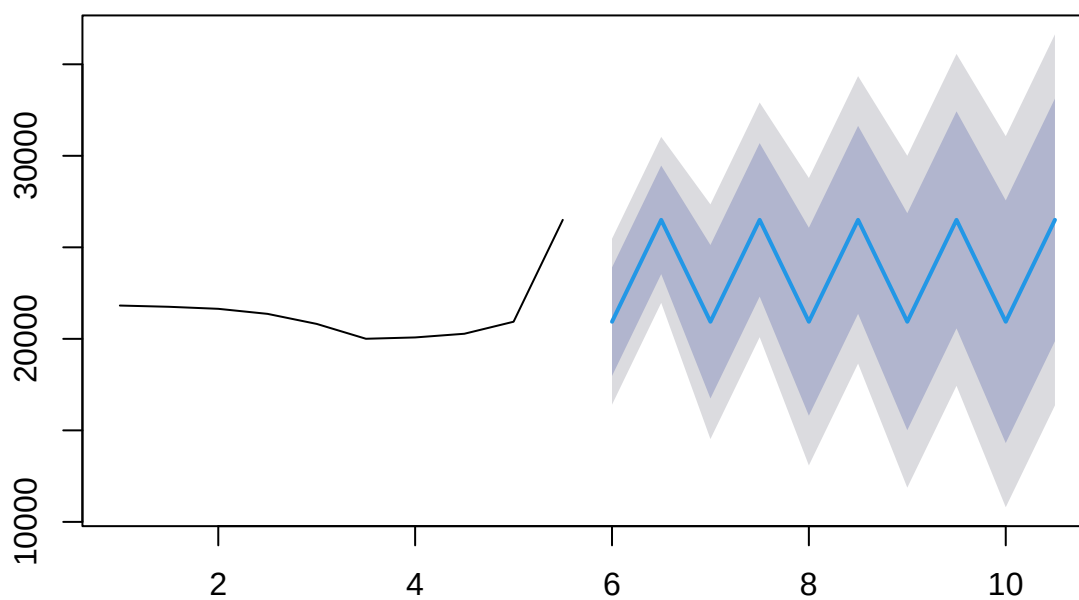
```
forecast <- forecast(fitcar, h = 10)
```

```
print(forecast)
```

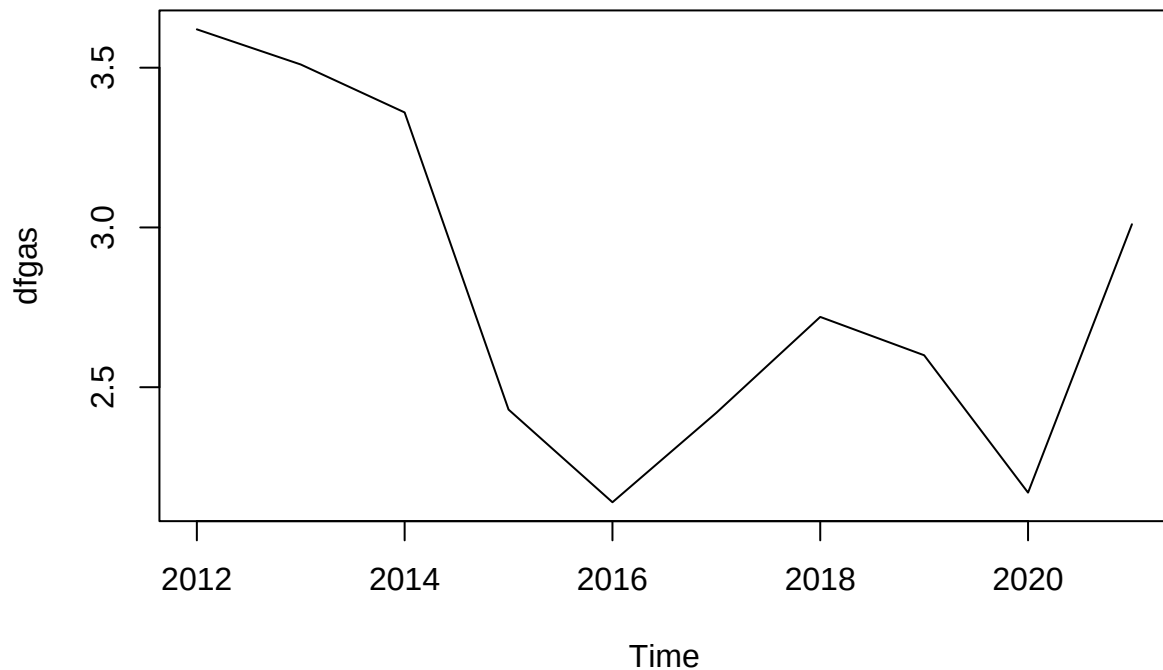
```
##      Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
## 2022      21518.4 19111.57 23925.23 17837.48 25199.32
## 2023      21518.4 19111.57 23925.23 17837.48 25199.32
## 2024      21518.4 19111.57 23925.23 17837.48 25199.32
## 2025      21518.4 19111.57 23925.23 17837.48 25199.32
## 2026      21518.4 19111.57 23925.23 17837.48 25199.32
## 2027      21518.4 19111.57 23925.23 17837.48 25199.32
## 2028      21518.4 19111.57 23925.23 17837.48 25199.32
## 2029      21518.4 19111.57 23925.23 17837.48 25199.32
## 2030      21518.4 19111.57 23925.23 17837.48 25199.32
## 2031      21518.4 19111.57 23925.23 17837.48 25199.32
```

```
plot(forecast(auto.arima(ts(dfcar,frequency=2),D=1),h=10))
```

Forecasts from ARIMA(0,0,0)(0,1,0)[2]



```
datagas <- data.frame(  
  Year = c(2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021),  
  USgas = c(3.62, 3.51, 3.36, 2.43, 2.14, 2.42, 2.72, 2.60, 2.17, 3.01)  
)  
  
dfgas <- ts(datagas$USgas, start = min(datagas$Year), end = max(datagas$Year), frequency = 1)  
  
plot(dfgas)
```



```
model <- auto.arima(dfgas, stepwise = FALSE, ic = "aic")
fitgas <- auto.arima(dfgas)
summary(fitgas)
```

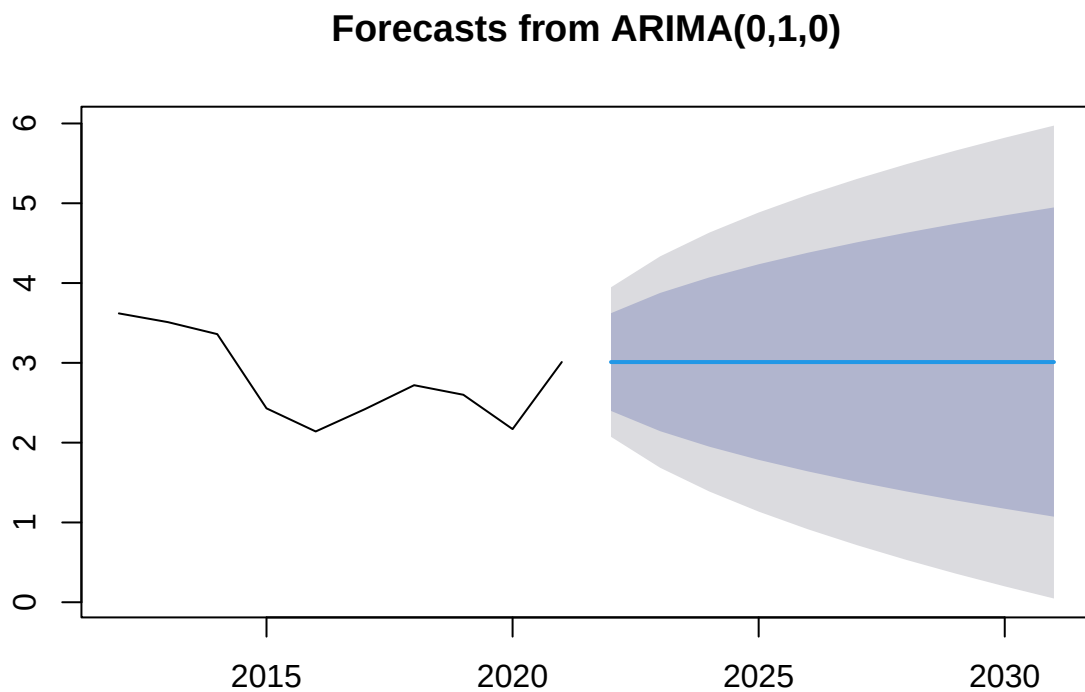
```
## Series: dfgas
## ARIMA(0,1,0)
##
## sigma^2 = 0.2285: log likelihood = -6.13
## AIC=14.26 AICc=14.83 BIC=14.45
##
## Training set error measures:
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set -0.060638 0.453532 0.345362 -3.324561 13.44589 0.9009443
##           ACF1
## Training set -0.004504763
```

```
forecast <- forecast(fitgas, h = 10)
print(forecast)
```

```
##      Point Forecast    Lo 80    Hi 80      Lo 95    Hi 95
## 2022           3.01 2.397335 3.622665 2.07301032 3.946990
```

```
## 2023      3.01 2.143561 3.876439 1.68489648 4.335104
## 2024      3.01 1.948834 4.071166 1.38708626 4.632914
## 2025      3.01 1.784671 4.235329 1.13602063 4.883979
## 2026      3.01 1.640040 4.379960 0.91482738 5.105173
## 2027      3.01 1.509284 4.510716 0.71485338 5.305147
## 2028      3.01 1.389042 4.630958 0.53095832 5.489042
## 2029      3.01 1.277123 4.742877 0.35979297 5.660207
## 2030      3.01 1.172006 4.847994 0.19903095 5.820969
## 2031      3.01 1.072584 4.947416 0.04697846 5.973022
```

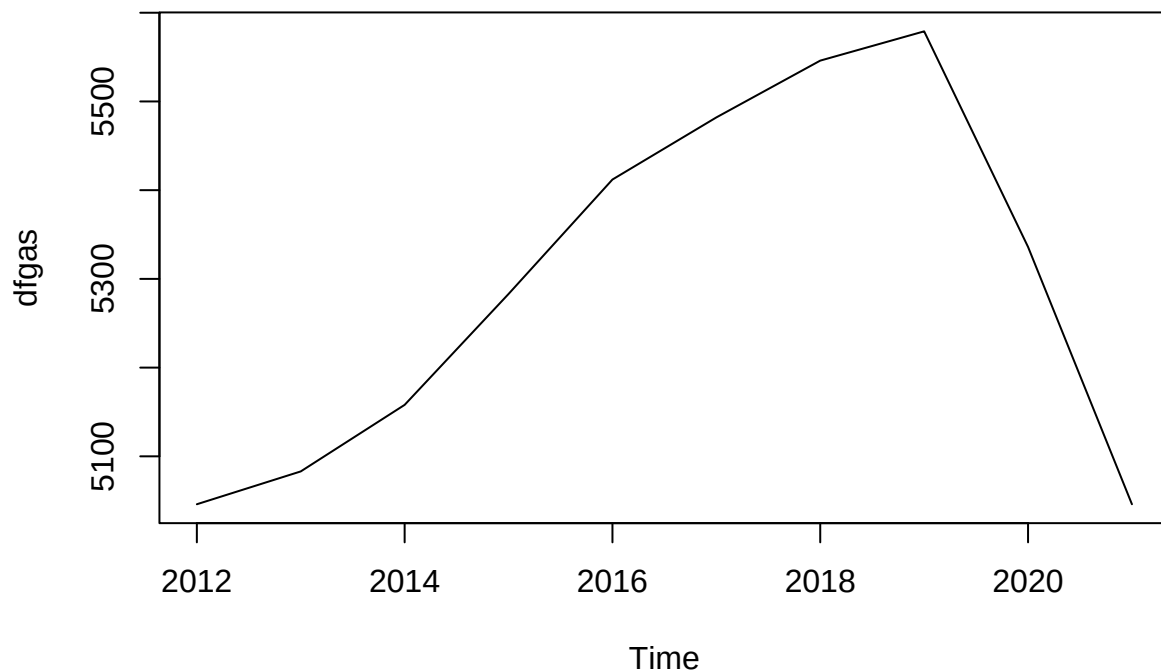
```
plot(forecast(fitgas, h = 10))
```



```
datapriv <- data.frame(
  Year = c(2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2021),
  USpriv = c(5046, 5083, 5158, 5283, 5412, 5482, 5546, 5579, 5336)
)

dfgas <- ts(datapriv$USpriv, start = min(datapriv$Year), end = max(datapriv$Year), frequency = 1)

plot(dfgas)
```



```
model <- auto.arima(dfgas, stepwise = FALSE, ic = "aic")
```

```
fitgas <- auto.arima(dfgas)
```

```
summary(fitgas)
```

```
## Series: dfgas
```

```
## ARIMA(1,0,0) with non-zero mean
```

```
##
```

```
## Coefficients:
```

```
##      ar1      mean
```

```
##      0.7549 5201.4357
```

```
## s.e. 0.1971 150.9966
```

```
##
```

```
## sigma^2 = 22836: log likelihood = -63.68
```

```
## AIC=133.35 AICc=137.35 BIC=134.26
```

```
##
```

```
## Training set error measures:
```

```
##      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
```

```
## Training set 17.06145 135.1621 119.1651 0.2571034 2.248118 1.006084 0.4666112
```

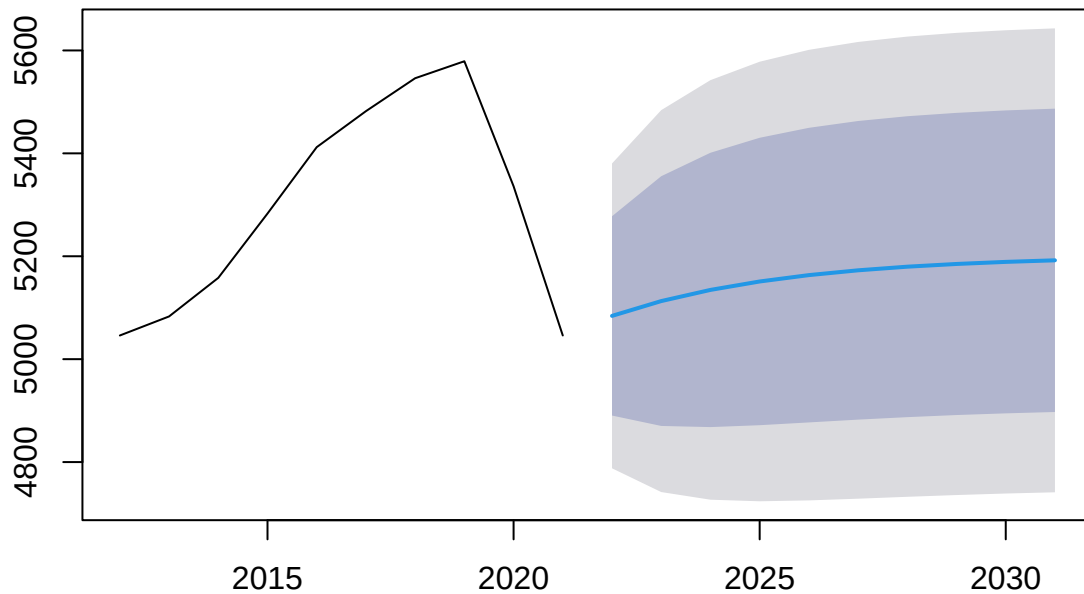
```
forecast <- forecast(fitgas, h = 10)
```

```
print(forecast)
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 2022	5084.095	4890.432	5277.757	4787.913	5380.276
## 2023	5112.853	4870.202	5355.504	4741.751	5483.956
## 2024	5134.563	4867.991	5401.135	4726.877	5542.250
## 2025	5150.953	4871.663	5430.242	4723.816	5578.089
## 2026	5163.325	4877.041	5449.610	4725.491	5601.159
## 2027	5172.665	4882.470	5462.861	4728.850	5616.481
## 2028	5179.717	4887.316	5472.118	4732.528	5626.905
## 2029	5185.040	4891.389	5478.690	4735.940	5634.139
## 2030	5189.058	4894.698	5483.418	4738.873	5639.243
## 2031	5192.092	4897.328	5486.856	4741.289	5642.894

```
plot(forecast(fitgas, h = 10))
```

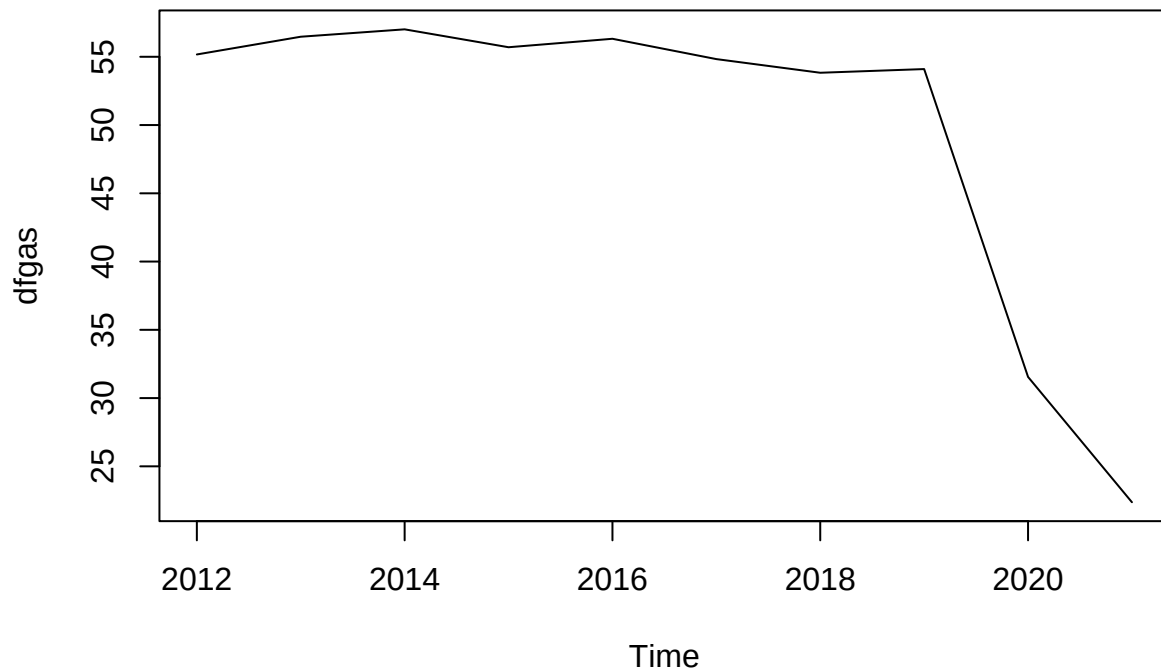
Forecasts from ARIMA(1,0,0) with non-zero mean



```
datapriv <- data.frame(
  Year = c(2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021),
  USpriv = c(55.17, 56.47, 57.01, 55.7, 56.32, 54.83, 53.83, 54.1, 31.55, 22.37)
)

dfgas <- ts(datapriv$USpriv, start = min(datapriv$Year), end = max(datapriv$Year), frequency = 1)

plot(dfgas)
```



```
model <- auto.arima(dfgas, stepwise = FALSE, ic = "aic")
fitgas <- auto.arima(dfgas)
summary(fitgas)
```

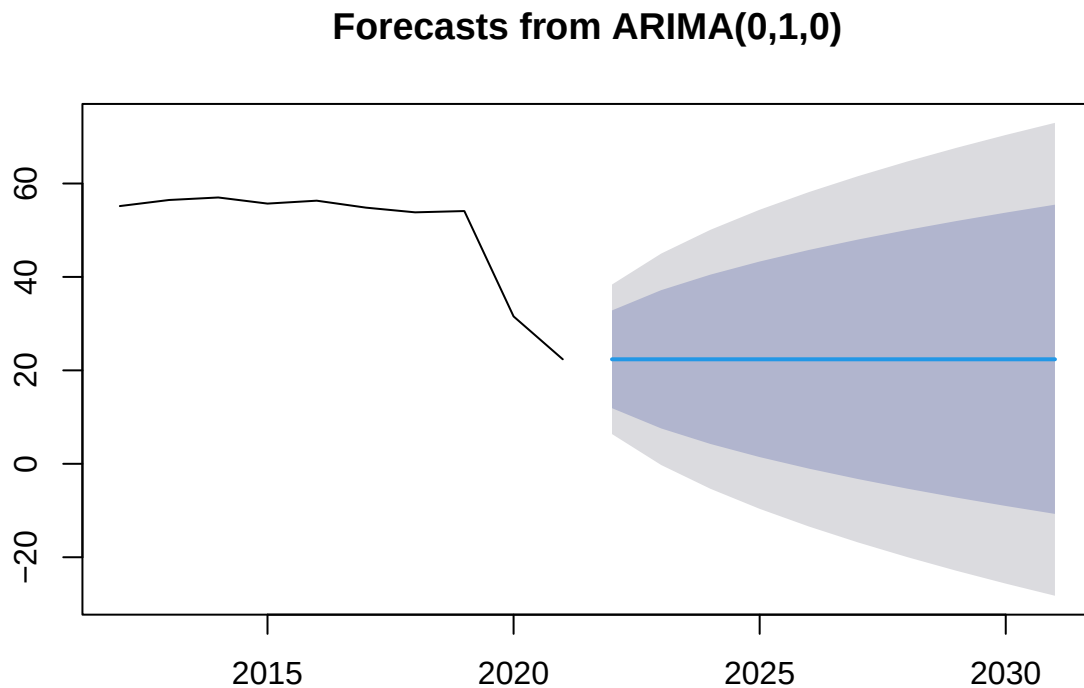
```
## Series: dfgas
## ARIMA(0,1,0)
##
## sigma^2 = 66.68: log likelihood = -31.67
## AIC=65.34 AICc=65.91 BIC=65.54
##
## Training set error measures:
##           ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
## Training set -3.274483 7.746955 3.831517 -11.44888 12.43873 0.9012978 0.2280347
```

```
forecast <- forecast(fitgas, h = 10)
print(forecast)
```

```
##      Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
## 2022          22.37  11.904841  32.83516   6.3649202  38.37508
## 2023          22.37   7.570030  37.16997  -0.2646009  45.00460
## 2024          22.37   4.243813  40.49619  -5.3516113  50.09161
```

```
## 2025      22.37   1.439682 43.30032  -9.6401595 54.38016
## 2026      22.37  -1.030807 45.77081 -13.4184463 58.15845
## 2027      22.37  -3.264300 48.00430 -16.8342787 61.57428
## 2028      22.37  -5.318209 50.05821 -19.9754608 64.71546
## 2029      22.37  -7.229940 51.96994 -22.8992017 67.63920
## 2030      22.37  -9.025477 53.76548 -25.6452393 70.38524
## 2031      22.37 -10.723739 55.46374 -28.2425062 72.98251
```

```
plot(forecast(fitgas, h = 10))
```



```
predictors <- read.csv("predict.csv")
```

```
## Warning in read.table(file = file, header = header, sep = sep, quote = quote, :
## incomplete final line found by readTableHeader on 'predict.csv'
```

```
predictors
```

```
##   Year  USAge  USCar  USGas PrivateMiles PublicMiles
## 1 2025   39.4  32910   4.24         4346         43.30
## 2 2026   39.6  28786   4.38         4121         45.80
## 3 2027   39.7  34351   4.51         3878         48.00
## 4 2028   39.8  30000   4.63         3635         50.06
```

```
predict(model1, predictors)
```

```
##           1           2           3           4
## 856.8813 863.1655 888.1523 852.3085
```

```
predict(modelLin, predictors)
```

```
##           1           2           3           4
## 14.85921 15.90499 16.18732 16.62025
```

```
predict(model2, predictors)
```

```
##           1           2           3           4
## 739.7401 754.9641 732.7598 710.0900
```

```
5.76*(14.85921 ^ 2)
```

```
## [1] 1271.786
```

```
5.76*(16.62025 ^ 2)
```

```
## [1] 1591.1
```

Model 1 predictions:

856.8813 in 2025 852.3085 in 2028

Linearized Model

1271.786 in 2025 1591.1 in 2028

Model 2 predictions:

739.7401 in 2025 710.0900 in 2028